

Module 2 — Bank Statements & Transaction Ledger (modules/banking)

Purpose

Import bank statements from Portuguese institutions (Millennium BCP, CGD, Santander, Novo Banco, Revolut, etc.), normalize them into the canonical **Transaction** ledger, auto-categorize on a 2-tier hierarchy, and reconcile with receipts, health claims, and utility bills. Detect recurring subscriptions, internal transfers, and statement gaps. Serve as the source-of-truth for household cash-flow analytics.

Actors & User Stories

- *As a user*, I import a CSV export from my Portuguese bank; the system auto-detects the column format, maps fields, and saves the import profile for next time—no remapping needed.
- *As a reviewer*, I confirm high-confidence receipt↔transaction matches and fix miscategorized rows in a unified queue.
- *As an analyst*, I query spend by 2-tier category, by tag, and per entity; internal transfers don't pollute income/expense totals.
- *As a household member*, I switch the entity filter to see just my own, my partner's, or our joint transactions — all of them are readable, the filter is for focus, not privacy.
- *As a maintainer*, I define categorization rules (e.g., "EDP → Electricity", regex patterns) and manage recurring charges.

Data Model

All monetary fields (**_eur** suffix) are **NUMERIC(10,2)** decimal EUR amounts (e.g. **19.99**), never minor-unit integers; UI displays them pt-PT-formatted (e.g. **19,99€**).

- **BankAccount**: **id**, **entity_id**, **institution**(MILLENNIUM_BCP|CGD|SANTANDER|NOVO_BANCO|REVOLUT|MANUAL|OTHER), **short_code** (e.g. AB, CR, REVO, MAN – the legacy “Source” code), **name**, **iban**, **currency**(EUR default), **account_kind**(CHECKING|SAVINGS|CREDIT|JOINT|MANUAL), **opening_balance_eur**, **is_deleted**, **locale_format**(pt_PT default).
- **Transaction** (canonical ledger, from D.3): **id**, **entity_id**, **account_id**, **booked_date** (legacy “Date”, the accounting/analysis date), **value_date** (legacy “DateOriginal”, original movement date – may fall in a different year), **amount_eur** (signed), **amount_net_eur** (absolute magnitude, derived), **currency**, **direction**(DEBIT|CREDIT|NONE), **description_raw**, **description_norm**, **merchant_id**, **category_l1_id**, **category_l2_id**, **category_nature**(EXPENSE|INCOME|TRANSFER|OTHER|NA – legacy “Category_type”), **bank_status**(PENDING|BOOKED|REVERSED|CANCELLED), **review_status**(TODO|DONE|WARN – human validation state, distinct from bank_status), **balance_after_eur**, **import_batch_id**, **split_group_fk** (legacy “FK”, groups split/related rows), **confidence**, **decision_reasons** JSONB, **notes** (legacy “Comments”, free text), **tags[]**, **raw_payload** JSONB.

- **RecurringCharge**: `id`, `entity_id`, `merchant_id`, `description_pattern`, `expected_amount_eur`, `cadence`(MONTHLY|QUARTERLY|ANNUAL|OTHER), `last_detected_at`, `status`(ACTIVE|PAUSED|ENDED), `last_amount_variance_pct`, `alert_threshold_pct`, `created_by`, `tags`[].
- **CsvMapping** (NEW): `id`, `bank_institution`, `name`, `column_map` JSONB{`date_idx`, `amount_idx`, `description_idx`, `balance_idx`, ...}, `delimiter`(COMMA|SEMICOLON|TAB), `decimal_sep`(COMMA|PERIOD), `date_format`(DD/MM/YYYY, ISO_8601, ...), `is_deleted`. Re-usable across imports.
- **TransferPair** (NEW, internal transfers): `id`, `from_transaction_id`, `to_transaction_id`, `paired_at`, `is_confirmed`, `created_by`. Prevents double-counting.
- **Rule** (NEW, auto-categorization): `id`, `entity_id`, `rule_type`(KEYWORD|REGEX|MERCHANT_EXACT), `pattern`, `category_l1_id`, `category_l2_id`, `priority`, `is_deleted`. Evaluated in order before merchant fallback.

Functional Requirements

- **FR-2.1 Statement Ingestion & Format Detection.** Parse CSV (comma/semicolon/tab), OFX (1.x, 2.0), and structured PDF exports. Auto-detect column structure, decimal separator (comma-decimal for PT), date format (DD/MM/YYYY). Normalize `booked_date/value_date`.
- **FR-2.2 2-Tier Categorization.** Assign each transaction to `Category (L1→L2)`. Auto-suggest via (1) user-defined `Rule` engine (keyword/regex), (2) learned merchant→category mapping, (3) description similarity fallback. Confidence scored per signal.
- **FR-2.3 Reconciliation & Linking.** Suggest/confirm `Link` edges to receipts (M1), health claims (M3), utility bills (M4), vehicles (M5). Support 1:N and N:1. Match on date (± 3 days configurable), amount (tolerance $\pm 5\%$ default), and merchant similarity.
- **FR-2.4 Contextual Tagging.** Apply multi-dimensional tags (`#vacation2026`, `#gifts`, `#daughter`) for cross-category aggregation; tags persist across linked entities.
- **FR-2.5 Recurring/Subscription Detection.** Identify recurring patterns by merchant and amount; alert when cadence shifts or amount deviates $>$ threshold (default 10%). Maintain `RecurringCharge` records.
- **FR-2.6 Internal Transfer Detection (NEW).** Detect transfers between accounts in the same household via matching amount and opposite direction within 1 business day. Create `TransferPair`; exclude both from income/expense totals. Surface for confirmation.
- **FR-2.7 Balance Continuity & Statement Gaps (NEW).** Validate `balance_after` chains within a statement period. Alert on gaps. Store statement metadata: opening/closing balance, period start/end.
- **FR-2.8 Import Profile Persistence (NEW).** Save column mappings, delimiter, date format, merchant normalizations as reusable `CsvMapping` per institution; select a saved profile on re-import.

Automation Rules (C.3 contract)

- **Categorization Engine:** Rules (keyword/regex) evaluated in priority order; unmatched fall to learned merchant mapping; low-confidence → review queue.
- **Reconciliation Engine:** Receipt↔transaction confidence combines date proximity (3-day window), amount tolerance ($\pm 5\%$), merchant similarity (≥ 0.80), OCR descriptor alignment. ≥ 0.85 auto-link; $0.60-0.85$ → review; < 0.60 flagged.

- **Internal Transfer Detector:** Two transactions in same household, opposite direction, same amount (within €1), booked_date within ± 1 day $\rightarrow$ propose **TransferPair**. Confirmed pairs excluded from cash-flow aggregates.
- **Recurring Pattern:** On each new transaction, scan history for same merchant/normalized description. ≥ 2 prior occurrences with regular cadence (± 3 days) $\rightarrow$ compute expected cadence/amount; deviation $>$ threshold $\rightarrow$ alert + **RecurringCharge** state change.
- **Statement Validation:** After parsing, check sequential **balance_after** for consistency; flag gaps and prompt user.

UI / Screens

- **Import Wizard:** upload $\rightarrow$ format auto-detect (CSV/OFX/PDF) $\rightarrow$ preview first 5 rows $\rightarrow$ column mapper (pre-populated from saved **CsvMapping**) $\rightarrow$ import confirmation + counts.
- **Ledger Grid** (TanStack Table): filterable/sortable by date, merchant, amount, category, entity, tags; inline category/tag edit; Pending/Booked badges; reconciliation status indicator.
- **Reconciliation Review Board:** split pane transaction $\leftrightarrow$ suggested matches with confidence; bulk actions; manual link; approval persists **Link** and advances linked-entity workflow state.
- **Recurring Charges Dashboard:** detected subscriptions (merchant, cadence, amount, last-seen, variance %); alert badge on threshold breach.
- **Statement Summary:** per import, opening/closing balance, period, count, cash-flow in/out, top categories, gaps/anomalies.
- **Rule Manager:** CRUD keyword/regex rules; drag to reorder priority; test on sample transactions before save.

API Surface

- **POST /accounts** · **GET /accounts** · **GET /accounts/{id}** · **PATCH /accounts/{id}** · **DELETE /accounts/{id}** (soft).
- **POST /transactions/import** (multipart: file + optional mapping_id, idempotent via Idempotency-Key) $\rightarrow$ { batch_id, counts, warnings[] }.
- **GET /transactions** (rich filters, pagination) · **GET /transactions/{id}** · **PATCH /transactions/{id}** · **POST /transactions/{id}/links**.
- **GET /recurring-charges** · **PATCH /recurring-charges/{id}**.
- **GET/POST /import-profiles** · **DELETE /import-profiles/{id}**.
- **CRUD /categorization-rules**.
- **POST /transactions/detect-transfers**.
- **GET /accounts/{id}/statement-summary**.

Analytics & KPIs

- Cash-flow: monthly income/expense by L1 category, net savings rate, trend.
- Category spend: L1 $\rightarrow$ L2 over time, vs budget (if defined).
- Tag rollups: aggregate by tag.
- Subscription spend: total monthly recurring, breakout by merchant, variance alerts.
- Uncategorized backlog: count of **category_l1_id IS NULL**, oldest first.
- Reconciliation rate: % linked to M1/M3/M4, by month.
- Balance health: opening vs closing per statement, unexplained gaps.
- Entity cash-flow: personal vs joint vs household net savings.

Edge Cases & Validation

- **Mixed-currency purchases (card abroad):** store both original currency and EUR-converted amount (statement rate); flag FX markup >2%.
- **Duplicate re-import safety:** dedupe key `hash(account_id, booked_date, amount_eur, description_norm)`; existing row → skip and log "already present".
- **Reversed/cancelled:** status tracks `PENDING → BOOKED/REVERSED`; reversals link back via `raw_payload.ref_transaction_id`.
- **Split transactions:** one bank row linked to multiple receipts via multiple `Link(type=RECONCILES)`.
- **Portuguese comma-decimal (1.234,56):** parser auto-detects via `CsvMapping.decimal_sep`; normalizes to a decimal EUR amount (e.g. `1234.56`), never minor units.
- **MB WAY / Multibanco references:** extract from description into `raw_payload.payment_method`; use for merchant/category hints.
- **Statement gaps:** flag in `ImportBatch` with gap range; user re-imports or confirms "not available".
- **Pending state changes:** allow `PENDING → BOOKED` without creating a duplicate.

Additional / Enriched Requirements

1. **Portuguese Bank Format Support (NEW):** codify CSV/OFX parsers for Millennium BCP, CGD, Santander, Novo Banco, Revolut (comma-decimal dates, accents, MB WAY identifiers); seed as `CsvMapping` templates.
2. **IBAN Validation (NEW):** validate Portuguese IBAN (PT + 24 digits); mask all but last 4 in UI.
3. **Internal Transfer Pair Semantics (NEW):** TransferPair as "zero-net-flow" for household; exclude from household income/expense; include in personal ledger only where the user is payer/payee.
4. **Rule Engine Extensibility (NEW):** user-ordered priority; **OR** logic for multiple keywords; test UI to preview matches on history.
5. **Statement Metadata Capture (NEW):** per-batch opening/closing balances; "verified with bank statement" flag + timestamp.
6. **Anomaly Alerting (NEW):** subscription amount change >threshold, recurring cadence gap, spend spike vs rolling-3-month average → Alert Engine.

Open Questions / Decisions

1. **Auto-link confidence threshold:** default 0.85; configurable per household/user.
2. **Internal transfer window:** default ±1 business day; consider 2–3 days for inter-bank delays.
3. **Recurring cadence tolerance:** default ±3 days; longer for annual charges.
4. **Statement gap handling:** auto-flag only (do not auto-reject).
5. **FX rates for multi-currency:** prefer statement-provided rate; historical lookup opt-in (privacy).

Definition of Done

- CSV/OFX parsers handle Portuguese bank formats; tests with anonymized seed examples.
- 2-tier categorization confidence ≥0.80 on sample; learns from confirmed corrections.
- Receipt↔transaction reconciliation ≥0.60 suggested; ≥0.85 auto-links (with confirmation option).
- Internal transfer detection excludes paired transfers from household cash-flow; tested.
- `CsvMapping` persistence; re-import uses saved profile.
- Rule engine prioritized; test UI; unit tests for evaluation order.
- Balance continuity validation; gaps detected & flagged; unit tests.

- Recurring detection; alerts on >10% variance; lifecycle tested.
- Ledger grid supports entity filtering, reconciliation status, inline edit.
- Reconciliation review board with bulk actions.
- Analytics: cash-flow, category spend, tag rollups, subscription spend, uncategorized backlog, reconciliation rate.
- API idempotent on re-import; RFC 9457 errors.
- ≥80% auto-process on sample; ≥80% coverage; pt-PT dates/decimal/currency on all UI.

Integration Contract

- **Exposes:** canonical **Transaction** ledger (D.3, shared source of truth); **Link** reconciliation edges to M1/M3/M4/M5; **RecurringCharge**; cash-flow aggregates for M6 and dashboards; entity-scoped transaction feeds.
- **Consumes:** **Merchant**, **Category** (L1→L2), **Tag**, **Entity**; **ImportBatch**, **ProcessingJob**, **ReviewTask**, **AuditLog**, **Document**; receipts/claims/bills/vehicles as link targets.
- **Triggers:** Reconciliation Engine, Alert Engine (anomalies/gaps/backlog), Analytics Engine.

Source-of-Truth Mapping (legacy Excel → model)

The household's current **Extratos_YYYY** sheets (~5,000 rows across 2024–2026) are the migration source. This mapping is authoritative for the CSV/legacy import path; every legacy column must round-trip.

Legacy column	Model target	Notes
Date	Transaction.booked_date	The accounting/analysis date (drives period rollups).
DateOriginal	Transaction.value_date	Original movement date; may be a different year than Date (cross-year re-accounting) — keep both, never collapse.
Date_Format , Year , Month , Date_ID , Date_text	<i>derived</i>	Do not store; compute (YYYYMMDD , YYYY_MM , month name) in queries/views.
Identity (Andre , Daniela , Andre_Daniela , Carolina)	Entity	Andre_Daniela → a two-member entity (the couple); Carolina → a dependent's single-member entity (no login). Confirms the M7 attribution model.
Source (AB , CR , REVO , MAN)	BankAccount.short_code / institution	MAN → institution=MANUAL

Legacy column	Model target	Notes
		(hand-entered, no statement).
Description	Transaction.description_raw (+ normalized description_norm)	
Amount	Transaction.amount_eur	Signed; negative = OUT/expense, positive = IN.
Amount_net	Transaction.amount_net_eur	Absolute magnitude; derived , never treated as a balance.
Category_1 / Category_2	category_l1_id / category_l2_id	2-tier hierarchy (banking).
Category_type (Despesas/Entrada/Outros/N/A)	Transaction.category_nature	EXPENSE / INCOME / OTHER / NA.
type (IN/OUT/N/A)	Transaction.direction	CREDIT / DEBIT / NONE (derived from sign).
Status (DONE/TODO/WARN)	Transaction.review_status	Human validation state — TODO →Review Queue, WARN →flagged/anomaly. Distinct from bank_status .
Comments	Transaction.notes	Free text; may encode manual rules — preserve verbatim.
Category_Aux	Tag	Event/context tag (e.g. a named trip).
Split	Link(SPLIT_OF) / payer_split	Split-movement info.
Is_Prendas	seed Tag #prendas	Recurring analytical flag (gifts).
Is_Puericultura	seed Tag #puericultura	Recurring analytical flag (childcare/child).
FK	Transaction.split_group_fk	Cross-reference key grouping split/related rows.

Aux sheet (controlled vocabularies: identities, sources, statuses, C1, and per-C1 C2 lists) seeds **Entity**, **BankAccount**, **Category** (L1→L2) and the review-status lookup — import it as the canonical starting vocabulary.